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29 April 1955
SUPERSEDING
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15 October 1946

MILITARY SPECIFICATION

GASKETS, ASBESTOS-METALLIC COVER, RUBBER CORE

All interested Bureaus of the Navy Department have concurred in the use of this specification.

1. SCOPE

1.1 This specification covers gaskets for use on hinged closures where flame resistance is required.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
HH-P-31 - Packing Material, Asbestos, Metallic Cloth, Sheet and Tape.
ZZ-R-601 - Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analysis).
LLL-B-631 - Boxes; Fiber Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid, (for Domestic Shipment).
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-P-120 - Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard.

NAVY DEPARTMENT
General Specifications for Inspection of Material,
Appendix I - Rubber and Synthetic Rubber; General Specification Methods of Testing.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
D-297-43T - Direct Determination of Rubber Hydrocarbon.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Construction. - Gaskets shall consist of a synthetic rubber core covered with two plies of asbestos metallic cloth or two successive asbestos metallic braids.

3.1.1 Asbestos metallic cloth. - The asbestos metallic cloth shall conform to Specification HH-P-31, except that it shall be flame-resistant and the cloth shall weigh not less than 3-1/2 pounds per square yard.

3.1.2 Asbestos metallic braids. - The asbestos metallic braids shall be constructed from two strands, each consisting of two asbestos yarns and two brass wires. The brass wire shall be 0.007 inch in diameter plus or minus 0.001 inch. The yarns shall contain not less than 90 percent of asbestos fiber and shall weigh not less than 1 pound per 1,000 yards. The braid shall be so constructed that there will be a minimum of 16 strands per inch in either direction, measured at right angles to the direction of the strands. The inner braid shall be bonded to the synthetic rubber core. The inner and outer braid shall be bonded with a noninflammable adhesive. The outer braid shall be treated to keep down lint in such a manner that it will not detract from flame resistance when the gasket is tested as specified in 4.5.5.

3.1.3 Rubber core. - The rubber core shall be a synthetic rubber material (with not more than 5 percent natural rubber hydrocarbon when tested as specified in 4.4.1) conforming to the following requirements:

3.1.3.1 Before aging:

(a) Tensile strength, minimum, pounds per square inch --- 1,000
(b) Ultimate elongation, minimum, percent ---------- 300
(c) Hardness:

Thickness, nominal Inch	Identation, minimum Millimeters
Less than 7/16	1.35
7/16 or more	1.40

3.1.3.2 After oven aging. - The synthetic rubber material, after having been subjected to the aging test specified in 4.4.2, shall meet the following requirements:

3.1.3.2.1 Tensile strength. - The tensile strength shall be not less than 75 percent of the tensile strength of unaged specimens.

3.1.3.2.2 Ultimate elongation. - The percent ultimate elongation shall be not less than 70 percent of the percent ultimate elongation of unaged specimens.

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3.1.3.2.3 Hardness. - The hardness determined by the method specified in 4.5.2 shall be not less than 1.15 millimeter indentation.

3.1.3.2.4 Compression set. - The compression set determined by the method specified in 4.4.3 shall be not more than 40 percent.

3.1.3.3 Delamination. - The synthetic rubber material shall show no evidence of separation into distinct layers or laminations, when tested in accordance with the procedure specified in 4.5.3.

3.1.3.4 Specific gravity. - The specific gravity of the synthetic rubber material shall be not greater than 1.40.

3.1.4 Adhesive. - The adhesive used between the inner and outer cover and between the inner cover and the rubber core shall be such that it will not be possible to strip the inner and outer asbestos metallic cloth or braid cleanly from each other or from the rubber without the use of solvents.

3.2 Dimensions. - Unless otherwise specified (see 6.1), the dimensions for the rubber core and the finished gasket shall be as shown in table I. The finished gasket shall be within a tolerance of plus 1/16 inch. No minus tolerance will be permitted. The minimum distance, measured diagonally across the corners of the gasket, shall be as shown in table I.

Table I - Dimensions.

Rubber core		Finished gasket			
Thickness (min.)	Width (min.)	Thickness (T) (min.)	Width (W) (min.)	Diagonal distance across corners (D) (min.)	Length
Inch	Inch	Inch	Inches	Inches	Feet
1/2	1/2	3/4	3/4	1-1/16	11-23
1/2	1	3/4	1-1/4	1-7/16	11-23

3.2.1 If dimensions other than those shown in table I are required, the finished gasket shall be of the same construction and the minimum distance, measured diagonally across the corners, shall be the square root of the sum of the squares of the thickness and the width of the finished gasket.

3.3 Flame and heat resistance. - The gasket shall satisfactorily pass the test specified in 4.5.5.

3.4 Workmanship. - Gaskets shall be free from defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. -

4.1.1 Lot. - All gaskets of the same size presented at one time shall be considered a lot for purposes of inspection and tests.

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4.1.2 Sampling for visual and dimensional inspection. - A random sample of gaskets shall be selected in accordance with table II from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Standard MIL-STD-105.

Table II - Sampling for visual and dimensional characteristics AQL (approx.) = 1.5 percent defective.

Number of gaskets in inspection lot	Number of gaskets in the sample
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

In addition, the manufacturer shall furnish with each lot a sample of the asbestos metallic cloth or of the asbestos metallic braid used in the manufacture of the lot of size deemed by the Government inspector to be sufficient for the weight determination specified in 4.2. Such samples of cloth and braid shall be guaranteed by the manufacturer to be from the material used for covering gaskets.

4.1.3 Sampling for tests. -

4.1.3.1 A random sample of gaskets shall be selected from each inspection lot in accordance with table III and shall be subjected to the tests specified in 4.4.1, 4.4.4, 4.5.2, 4.5.3, 4.5.4, and 4.5.5.

Table III - Sampling for tests.

Number of gaskets in inspection lot	Number of gaskets in the sample
25 and under	2
26 to 110	3
111 to 300	5
301 to 500	7
501 and over	8

4.1.3.2 Sampling for tensile strength and elongation tests. - For the tests specified in 4.5.1 the contractor shall furnish two sample pieces 6 inches long by 6 inches wide by 0.075 inch thick. These pieces shall be identical in composition and equivalent in cure; and prepared at the same time from material used in the lot of gaskets offered for delivery.

4.1.3.3 Sampling for compression set tests. - The contractor shall furnish one sample piece approximately 4 inches long by 1-1/2 inches wide and 1/2 inch in thickness. This sample piece shall be identical in composition and equivalent in cure; and prepared at the same time from material used in the lot of gaskets offered for delivery.

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4.2 Lot acceptance inspection. - Each of the sample gaskets selected in accordance with 4.1.2 shall be visually and dimensionally inspected by the Government inspector to verify conformance with this specification. The inspection shall include determination of the weight of the samples of asbestos metallic cloth and braid. Any gasket in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective gaskets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming gaskets. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

4.3 Classification of tests. - Classification of tests shall be as follows:

(a) Production check tests.
(b) Lot acceptance tests.

4.4 Production check tests[1]. - Production check tests shall be conducted on the first lot of material offered for delivery under a contract or order and thenceforth samples shall be furnished for every 10 lots. Production check tests shall be made at the place of manufacture by or under the supervision of the Government inspector, if facilities are available; otherwise production check tests shall be made at a Government laboratory designated by the bureau or agency concerned. Samples selected in accordance with 4.1.3.2 and 4.1.3.3 shall be subjected to the following tests and also to those specified in 4.5.1 (before and after aging), and 4.5.2 (before and after aging).

4.4.1 Determination of natural rubber hydrocarbon. - The percent of rubber hydrocarbon shall be tested as specified in the Direct Determination of Rubber Hydrocarbon of ASTM Method D-297-43T.

4.4.2 Aging test. - Specimens for tensile strength test, elongation test, hardness test and for compression set test shall be given an accelerated aging test by subjecting them to dry circulating air at a temperature of 194 ± 2°F. (90 ± 1.1°C.), for 46 hours. Final determination of aged tensile strength test specimens shall be made not less than 20 hours or more than 48 hours after removal from the oven. Tensile strength and elongation tests on unaged specimens shall be made immediately prior to, and on the same machine as, the tensile strength and elongation tests on the aged specimens.

4.4.3 Compression set test. -

4.4.3.1 Apparatus. - The compression device shall consist of two or more flat steel plates between the parallel faces of which the specimens may be compressed. Steel spacers 0.300 inch thick shall be placed on each side of the rubber specimens to control their thickness to a compression of 40 percent. Four bolts shall pass through holes at the corners of the plates for compressing and holding the specimens.

4.4.3.2 Test specimens. - The test specimens shall be 1.129 inches in diameter by 1/2 inch thick, cut and buffed to within a permissible variation of plus or minus 0.005 inch in any dimension.

4.4.3.3 Measurements. - The thickness before and after the compression set test shall be measured to the nearest 0.001 inch at the center of each specimen on a dial gage graduated in thousandths of an inch, having a spherical foot 1/4 inch in diameter exerting a total pressure of 3 ounces, ±0.1 ounce.

4.4.3.4 Procedure. - Initial thickness measurements and specimen loading operations shall be performed at room temperature. Not less than 5 minutes nor more than 30 minutes after completing the loading operation, the specimen, clamped to 40 percent compression, shall be subjected to the conditions specified in 4.4.2. The compression set clamps shall be taken from the oven and the test specimen immediately removed from the clamps. The specimen shall be allowed to cool at room temperature, on a wooden surface, for 30 minutes before the final thickness is measured. A duplicate test shall be run and shall agree within 5 percent with the first test.

[1]Unless otherwise specified in the test method, no tests shall be conducted prior to a conditioning period of 4 hours at room temperature. Sample preparation may be undertaken without regard to this time interval.

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4.4.3.4.1 Computation. - The compression set shall be computed in accordance with the following formula:

$$\text{Compression set (percent)} = \frac{t_o - t_1}{t_o - t_s} \times 100$$

Where:

t_o = the original thickness of the test specimen.
t_1 = the thickness of the test specimen at the time specified after removal from the clamp.
t_s = the thickness of the steel spacers.

4.4.4 Specific gravity. - The specific gravity shall be determined by using the hydrostatic weighing method, or other method satisfactory to the bureau or agency concerned, as specified in Specification ZZ-R-601.

4.4.5 Action in case of failure. - If any one of the samples is found not to be in conformance with 3.1.3, the lot which it represents shall be rejected. Furthermore, additional samples shall be selected from (or to represent) each subsequent lot and shall be subjected to the tests specified in 4.4. A lot shall then be accepted only upon satisfactory completion of the tests by all the samples selected to represent the lot. This additional testing shall be discontinued when 4 successive lots have passed the test or tests.

4.5 Lot acceptance tests[1]. - Lot acceptance tests shall be conducted at the place of manufacture if facilities are available. Otherwise, lot acceptance tests shall be conducted at a Government laboratory designated by the bureau or agency concerned.

4.5.1 Tensile strength, elongation tests. - Methods of tests and type D test specimens for tensile strength and elongation of Specification ZZ-R-601 shall be used.

4.5.2 Hardness test. - The hardness shall be determined using a plunger of 3.2 millimeter diameter and the test method specified in Specification ZZ-R-601 and Appendix I of General Specifications for Inspection of Material. Samples of material less than 1/4 inch thick shall be superimposed to a thickness of not less than 1/4 inch or greater than 1/2 inch before making the hardness determination.

4.5.3 Delamination test. - The delamination test shall be in accordance with the methods specified in Appendix I of General Specifications for Inspection of Material.

4.5.4 Asbestos cloth and asbestos metallic braid. - Tests on asbestos cloth and asbestos metallic braid shall be in accordance with Specification HH-P-31 (see 3.1.1).

4.5.5 Flame and heat resistance of gaskets. - A specimen of the completed gasket shall be held for 15 seconds with one face across the flame of a Tirrill type burner. The lower side of the gasket shall be held even with the tip of the blue cone of the flame. The temperature of the flame, measured with a pyrometer, shall be approximately 1,600°F. at the tip of the blue cone. If a pyrometer is not available for measuring the temperature, the tip of the blue cone of the flame shall be not less than 1 inch or more than 1-1/2 inches high. The specimen, when cut open at the exposed section, shall not show any indication of melting of the rubber core, nor shall the adhesive compound have been consumed. In addition, a specimen of the gasket shall be held, for not more than 60 seconds, over a 1,600°F. flame, as specified herein, until the rubber core possibly ignites or bursts into flame. The gasket, if burning, shall not continue to blaze when removed from the burner flame for more than 60 seconds. Cloth or braid covering shall hold its shape and texture.

[1]Unless otherwise specified in the test method, no tests shall be conducted prior to a conditioning period of 4 hours at room temperature. Sample preparation may be undertaken without regard to this time interval.

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4.5.6 Action in case of failure. - Failure of any one of the samples selected in accordance with 4.1.3 to pass any one of the tests specified in 4.5 shall be cause for rejection of the lot represented by the sample. Subject to the approval of the Government inspector, the manufacturer may rework each defective item in a rejected lot and may then resubmit the lot for Government inspection. In this case, the lot shall be resampled by the Government inspector in accordance with 4.1.2 and 4.1.3 and shall be subjected to inspection and tests specified in 4.2 and 4.5. Material rejected after retests shall not be resubmitted for Government inspection without the specific approval of the bureau or agency concerned.

4.6 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5 PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 For domestic shipment, immediate use. - The gaskets shall be packaged in accordance with commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment. - Unless otherwise specified in the contract or order, the gaskets shall be individually packaged in folding cartons or set-up boxes conforming to Specifications JAN-P-120 and PPP-B-676, respectively.

5.2 Packing. -

5.2.1 For domestic shipment, immediate use. - The gaskets shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2.2 For domestic shipment and storage. - Gaskets, packaged as specified in 5.1.2, shall be packed in corrugated or solid fiberboard boxes conforming to the special requirements of Specifications LLL-B-631 and LLL-B-636, respectively.

5.2.3 For overseas shipment. - Gaskets, packaged as specified in 5.1.2, shall be packed in fiberboard boxes conforming to Specification JAN-P-108.

5.3 Marking for shipment. - In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Dimensions required when not as specified in table I (see 3.2).
(c) Whether for domestic shipment, immediate use; domestic shipment and storage; or overseas shipment (see 5.1 and 5.2).

6.2 Gaskets should not be lubricated with graphite or other material.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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